AWS KMS — Important Interview Questions & Answers

1.What is AWS KMS?
AWS KMS stands for Key Management Service. It is an AWS service used to create, manage, and control encryption keys used to protect data.
Simple ga gurthupettuko:
KMS ante AWS lo encryption keys ni create and manage cheyyadaniki use chese service.
Eng: AWS KMS is a service used to create and manage encryption keys.

2.What is a Customer Managed Key?
A Customer Managed Key is an encryption key that we create and manage in AWS KMS. We have control over its policies, permissions, rotation, and lifecycle.
Simple ga gurthupettuko:
Customer Managed Key ante maname create chesi manage chese KMS key.
Eng: A Customer Managed Key is a KMS key that we create and manage.

3.What is an AWS Managed Key?
An AWS Managed Key is a KMS key created and managed by AWS on behalf of an AWS service. It is used by supported AWS services for encryption.
Simple ga gurthupettuko:
AWS Managed Key ante AWS service kosam AWS automatic ga manage chese KMS key.
Eng: An AWS Managed Key is a KMS key managed by AWS for an AWS service.

4.What is the difference between Customer Managed Key and AWS Managed Key?
A Customer Managed Key is created and controlled by the customer, while an AWS Managed Key is created and managed by AWS for a supported service.
Simple ga gurthupettuko:
Customer Managed Key manam control chestham, AWS Managed Key ni AWS manage chestundi.
Eng: Customer Managed Keys are managed by customers, while AWS Managed Keys are managed by AWS.

5.What is a KMS Key Policy?
A Key Policy is a resource-based policy that controls access to a KMS key. It defines which users, roles, or AWS accounts can use or manage the key.
Simple ga gurthupettuko:
Key Policy ante KMS key ni evaru use or manage cheyyalo decide chese policy.
Eng: A Key Policy controls who can use or manage a KMS key.

6.What are KMS Grants?
KMS Grants provide a way to give permissions to AWS principals to use a KMS key. They are useful when temporary or programmatic permissions are required.
Simple ga gurthupettuko:
KMS Grant ante KMS key ni use cheyyadaniki specific permissions ivvadam.
Eng: KMS Grants provide permissions to use a KMS key.

7.What is Key Rotation?
Key Rotation means periodically generating new cryptographic key material for a KMS key. It helps improve security while allowing the key to continue being used.
Simple ga gurthupettuko:
Key Rotation ante security kosam KMS key material ni periodically rotate cheyyadam.
Eng: Key Rotation periodically changes the cryptographic key material for a KMS key.

8.What is a KMS Key Alias?
A KMS Key Alias is a friendly name for a KMS key. It makes it easier to identify and reference a key instead of using its long key ID.
Simple ga gurthupettuko:
Key Alias ante KMS key ki easy ga gurthupettukune friendly name.
Eng: A KMS Key Alias is a friendly name used to identify a KMS key.

9.Why do we use AWS KMS?
We use AWS KMS to securely create and manage encryption keys and control who can use those keys. It is commonly used to encrypt data in AWS services.

Simple ga gurthupettuko:
KMS ni data encryption kosam keys secure ga create and manage cheyyadaniki use chestham.
Eng: AWS KMS is used to manage encryption keys and protect data.

10.What is Encryption?
Encryption is the process of converting readable data into an unreadable format using a cryptographic key. The data can be converted back to its original form using the appropriate key.
Simple ga gurthupettuko:
Encryption ante readable data ni key use chesi secure unreadable format lo convert cheyyadam.
Eng: Encryption converts readable data into a protected format using a key.

11.What is Decryption?
Decryption is the process of converting encrypted data back into its original readable form using the required cryptographic key.
Simple ga gurthupettuko:
Decryption ante encrypted data ni key use chesi original readable data ga convert cheyyadam.
Eng: Decryption converts encrypted data back into readable data.

12.What is the difference between KMS and IAM?
KMS is used to create and manage encryption keys, while IAM is used to manage identities and permissions for AWS resources.
Simple ga gurthupettuko:
KMS encryption keys kosam, IAM users and permissions kosam use chestham.
Eng: KMS manages encryption keys, while IAM manages identities and permissions.

13.Can we use a KMS Key with Amazon S3?
Yes. KMS keys can be used with Amazon S3 for server-side encryption. This allows us to control encryption keys and permissions using AWS KMS.
Simple ga gurthupettuko:
S3 data ni encrypt cheyyadaniki KMS key use cheyyachu.
Eng: KMS keys can be used to encrypt data stored in Amazon S3.

14.Can we use KMS with EBS?
Yes. AWS KMS keys can be used to encrypt EBS volumes and snapshots. This helps protect data stored on Amazon EBS.
Simple ga gurthupettuko:
EBS volumes and snapshots ni encrypt cheyyadaniki KMS use cheyyachu.
Eng: KMS keys can be used to encrypt EBS volumes and snapshots.

15.What happens if we disable a KMS key?
When a KMS key is disabled, it cannot be used for cryptographic operations such as encryption or decryption. Applications depending on that key may be unable to access encrypted data until the key is enabled again.
Simple ga gurthupettuko:
KMS key disable chesthe encryption or decryption operations work avvavu.
Eng: A disabled KMS key cannot be used for cryptographic operations.

16.What happens if we delete a KMS key?
Deleting a KMS key can make data encrypted with that key inaccessible if no other valid decryption mechanism is available. AWS provides a waiting period for customer managed key deletion so that accidental deletion can be prevented.
Simple ga gurthupettuko:
KMS key delete chesthe aa key tho encrypt chesina data ni decrypt cheyyadam problem avvachu.
Eng: Deleting a KMS key can make data encrypted with that key inaccessible.

17.What is the difference between Key Policy and IAM Policy in KMS?
A Key Policy is attached directly to a KMS key and controls access to that key. IAM policies can also grant permissions for KMS actions when the key policy allows the account to use IAM policies for access control.
Simple ga gurthupettuko:
Key Policy direct ga KMS key access ni control chestundi, IAM Policy user or role permissions ni control chestundi.
Eng: Key Policies control access to KMS keys, while IAM Policies control identity permissions.

18.How would you give an EC2 instance access to a KMS key?
I would attach an IAM Role to the EC2 instance and provide the required KMS permissions through IAM and the KMS key policy.
Simple ga gurthupettuko:

EC2 ki IAM Role attach chesi required KMS permissions ivvachu.
Eng: I would use an IAM Role with the required KMS permissions for EC2.

19.How would you troubleshoot a KMS Access Denied error?
I would check the IAM policy, KMS key policy, grants, key status, and whether the correct user or role is being used. I would also verify that the required KMS action is allowed.
Simple ga gurthupettuko:
KMS Access Denied vaste IAM Policy → Key Policy → Grants → Key Status → User/Role check chestham.
Eng: For a KMS Access Denied error, I would check IAM policies, key policies, grants, and key status.

20.What is the most important thing to remember about AWS KMS?
AWS KMS is used to create, manage, and control encryption keys in AWS. The important concepts are Customer Managed Keys, AWS Managed Keys, Key Policies, Grants, Key Rotation, and Key Aliases.
Simple ga gurthupettuko:
KMS ante AWS lo encryption keys ni securely create, manage, and control cheyyadaniki use chese service.
Eng: AWS KMS is used to securely create, manage, and control encryption keys.